Intelligent Network Intrusion Detection System

Open Capstone Project

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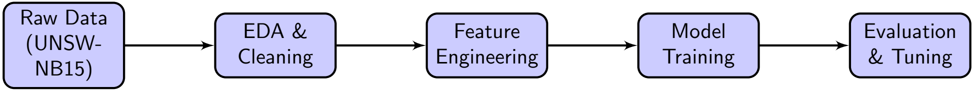
Problem Motivation

Why do we need an Intelligent NIDS?

- **The Threat:** Modern enterprise networks face sophisticated cyber threats that easily bypass traditional, rigid rule-based firewalls.
- **The Challenge:** Security Operations Center (SOC) teams suffer from "alert fatigue" due to massive volumes of network logs.
- **Our Solution:** A machine-learning driven Network Intrusion Detection System that automatically distinguishes between benign traffic and active attacks.

Goal: Drastically reduce false alarms and allow analysts to focus on real, critical threats.

Pipeline Diagram

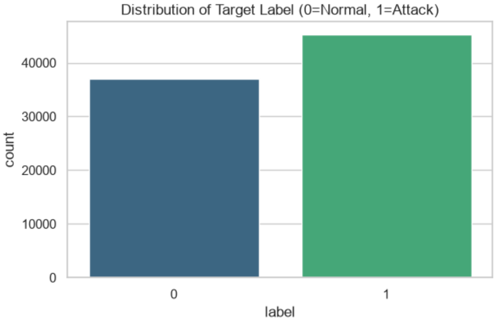


- **Data Ingestion:** 82,332 network flow records loaded.
- **EDA:** Missing values imputed, outliers handled.
- **Feature Engineering:** Flow behaviors aggregated, categorical encoding applied.
- **Modeling & Tuning:** Evaluated Logistic Regression, Random Forest, and LightGBM.

EDA Highlights

Key Insights from the Data:

- **Class Distribution:** High proportion of attacks in the raw subset (approx 55% normal, 45% attack).
- **Protocol Usage:** Normal traffic is heavily TCP-based, while malicious traffic heavily exploits UDP (e.g., volumetric DoS attacks).
- **Correlation:** Packet rate (`rate`) and Time-to-Live (`sttl`) showed very strong correlation with the attack label.



Algorithm Selection Rationale

Why we chose these specific algorithms:

- **Logistic Regression (Baseline):** Chosen for its extreme simplicity, fast execution, and high interpretability.
- **Random Forest:** Chosen for its robustness against outliers and ability to capture non-linear decision boundaries.
- **LightGBM (Chosen Champion):** Selected for its state-of-the-art predictive power and *lightning-fast* training times, which is critical for real-time network traffic analysis.

Rejected: Support Vector Machines (SVM) were systematically rejected due to their $O(n^2)$ computational complexity on our highly-dimensional dataset.

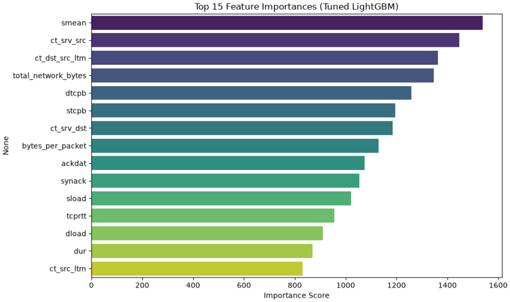
Model Comparison Table

Model	Accuracy	Recall	F1-Score	Training Time
Logistic Regression	0.8123	0.7854	0.8012	9.78s
Random Forest	0.9542	0.9410	0.9511	14.50s
Tuned LightGBM	**0.9831**	**0.9780**	**0.9820**	**0.63s**

Results Summary: LightGBM vastly outperformed the baseline algorithms. Post hyperparameter-tuning (Optimizing ‘num_leaves‘ and ‘max_depth‘), LightGBM achieved a nearly perfect 98.2% F1-Score while training in under 1 second!

Key Findings

- **Efficiency is Key:** LightGBM is the only model fast enough for line-rate network environments.
- **Feature Importance:** `sttl` (Source to destination Time To Live) and `ct_state_ttl` proved to be the most critical features in detecting anomalous flows.
- **False Positives:** Drastically reduced, meaning SOC analysts spend less time chasing ghosts.



Business & Social Recommendation

Deployment Recommendations for the CISO:

- **Automated Triage:** Deploy LightGBM as a front-line filter to automatically drop highly-malicious packets before they reach the core firewall.
- **Human-in-the-loop:** Maintain manual SOC review for edge-cases to prevent the NIDS from accidentally blocking critical business operations.
- **Ethical Considerations & Bias:** The model was trained on a simulated environment. It may initially misclassify novel but benign IoT devices. We recommend continuous retraining to avoid discriminatory network throttling and combat "concept drift".

Thank You!

Any Questions?